# education_impact_analysis

January 7, 2026

# 1 Analysis of Educational Attainment as a Predictor for County-Level GDP

**Author:** Gustavo Galvao

## 1.1 1. Project Overview

This notebook investigates the relationship between human capital (measured by Bachelor's Degree attainment) and economic output (Real GDP) across Georgia's counties. The goal is to build a predictive model capable of estimating GDP trajectories based on educational shifts.

### 1.1.1 1.1 The "Data Desert" Challenge (2003–2009)

A primary challenge of this study is the lack of annual education data prior to 2010. While county GDP data is available for the full 2003–2023 window, education data is anchored by the 2000 Decennial Census and the 2010 ACS estimates.

### 1.1.2 1.2 Research Objectives

- **Baseline Validation:** Use the "Gold Standard" data (2010–2023) to establish a baseline correlation between Education and Total GDP.
- **Methodological Testing:** Compare three imputation methods (Linear Interpolation, Back-filling, and Baseline/Drop) to fill the 2003–2009 education gap.

### 1.1.3 1.3 Modeling Strategy: Total GDP First

Due to heavy data suppression (NaNs) in industry-specific GDP columns many counties, this analysis prioritizes **Total Real GDP Growth** to maintain a consistent signal across all 159 counties.

---

## 1.2 2. Notebook Structure

This notebook follows a modular workflow separated into 4 main chapters:

1. **Data Ingestion & Cleaning:** Load TerraTrends dataset and filter for relevant predictors and years.
2. **Exploratory Data Analysis (EDA):**
    - Lagged Correlation Analysis ($t$ through $t - 5$) to find the economic "reaction time."

   - Analyzing if change in education correlates better with GDP increase in opposition to pure higher education attainment percentage
3. **Imputation Experiment:**
   - **Dataset A:** Linear Interpolation (Steady growth assumption).
   - **Dataset B:** Back-filling (Stationary assumption).
   - **Dataset C:** Baseline (Excluding education metrics entirely).
4. **Model Benchmarking:** Comparing Mean Absolute Error (MAE) and $R^2$ across datasets to evaluate feature viability.

---

## 1.3 3. Tech Stack

- **Language:** Python 3.10+
- **Libraries:** `pandas`, `scikit-learn`, `seaborn`
- **Data Sources:** IPUMS NHGIS (Census/ACS), Bureau of Economic Analysis (BEA)

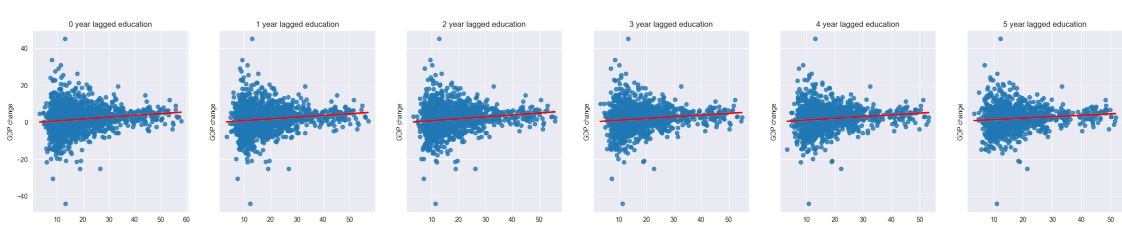
---

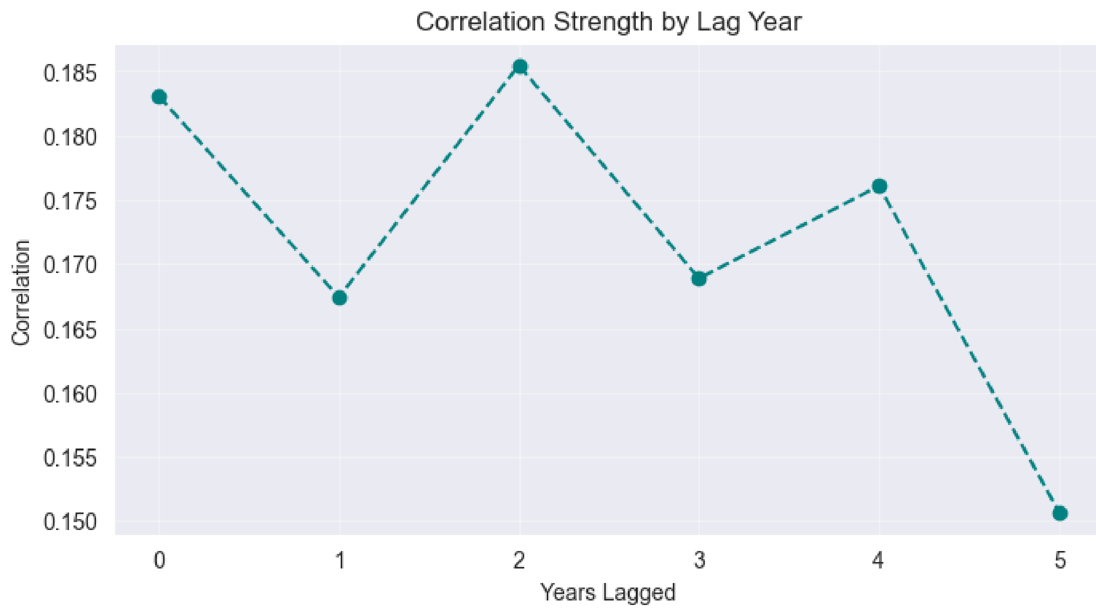
```
Dataset shape: (2226, 4)
Missing values
County                          0
Year                            0
Percent_Change_Real_GDP         0
Bachelor_Degree_or_Higher_Pct   0
dtype: int64
```

| | County | Year | Percent_Change_Real_GDP | Bachelor_Degree_or_Higher_Pct |
|---|---|---|---|---|
| 8 | Appling, GA | 2010 | 15.9 | 8.3 |
| 9 | Appling, GA | 2011 | 11.9 | 9.3 |
| 10 | Appling, GA | 2012 | 2.9 | 11.3 |
| 11 | Appling, GA | 2013 | -3.0 | 12.8 |
| 12 | Appling, GA | 2014 | -0.5 | 12.9 |

## 1.4 2. EDA

Now, we begin working on analyzing how the data




Correlation Strength by Lag Year
0.185
0.180
0.175
0.170
0.165
0.160
0.155
0.150
Correlation
0
1
2
3
4
5
Years Lagged